

PROFESSIONAL SUMMARY

- Excellent analytical skills, networking, and management skills
- Excellent reasoning and writing skills with proven peer-reviewed publications
- Experience in setting up analytical pipelines, resource planning, and creating interactive data visualizations to draw insights from large sets of data
- Experience in implementing statistical algorithms and methods for big data analytics and data mining in NGS analysis
- Over 10 years of work experience in clinical and biomedical life sciences
- MS/Ph.D. in integrative life sciences with extensive training in precision medicine for cancer, bioinformatics, stem cell biology, biostatistics, and psychiatry research

WORK EXPERIENCE

Research Scientist, OSU, Advanced Molecular Diagnostics, OSUMC, 2023 - Present
OH-USA

- Spearhead the development and execution of advanced next-generation sequencing (NGS) data analysis pipelines, enhancing the precision and efficiency of delivering the analysis to collaborators and clients.
- Engineer comprehensive workflow solutions to optimize data acquisition, analytical processing, and interactive reporting mechanisms within the COVID-19 surveillance initiative. [Demo](#)
- Foster collaborative partnerships with scientific peers, providing expert analysis and support for specialized sequencing projects, including targeted DNA sequencing, RNA-seq, and single-cell RNA-seq (sc-RNA-seq).
- Serve as a key contributor in the advancement and diversification of sequencing capabilities at AMD, driving innovation in research methodologies and clinical applications, particularly in oncology and integrated multi-omics studies.

Contractor (Bioinformatics), OSU, Advanced Molecular Diagnostics, 2022 - 2023
OSUMC, OH-USA

- Improved and automated the process of COVID-19 viral sequence alignment, analysis, and the submission of sequence data to the consortium.

- Investigate other viral genome alignment pipelines and reporting systems and compare the outputs with our pipeline
- Implement clinical testing reporting system to Ohio Department of Health while collaborating with Ohio partners and develop user interface in Shiny for analysis and reporting pipeline. [Demo](#)

Research Scientist, OSU, Research IT-Data Science Group, Office of 2019 - 2022
Research, OH-USA

- Build automated data processing pipelines for developing analytical dashboards using R - Shiny, Tableau, R-markdown reports in state wide [NIH HEALing Communities Study](#)
- Data analytics involving opioids and opioid use disorder data collected over multiple agencies in Ohio to understand and reduce the substance use.
- Developed public facing State of Ohio Integrated Behavioral Health [Tableau dashboard](#) that assist in improving behavioral health awareness and decision-making working with stakeholders in involved in Recovery Ohio project.
- [Building interactive visual network](#), gene-gene interaction pathway networks using R shiny on large scale data collected on cell death and drug-gene interaction pathways

2017 - 2019

Research Scientist, OSU, Biomedical Informatics, OH-USA

- Supporting independent PI's or collaborators at OSU research departments in interpreting clinical or research NGS data in the context of their specific research questions
- Running existing NGS pipelines or creating custom pipelines at the Bioinformatics shared resources (BISR) and creating QA/QC reports across different types of raw NGS data (RNA seq, Exome, ChipSeq)
- Responsible for the development of interactive data analytical pipelines for the analysis of RNA counts data or VCF files
- Extensively worked in collaboration with Ohio Supercomputer (OSC) group to create a self-contained user-friendly analytical web application using R-shiny to securely deliver NGS reports to clinicians/researchers.

Bioinformatics Database Analyst, VCU Health, VA- USA

2015 - 2017

- Synthesize metadata data from various sources in context to ACMG clinical classification guidelines of DNA sequence variants and integrate the data as part of our in-house visual based application Next Generation Sequencing Interface (NeGeSi).
- Significantly contributed in designing and developing NeGeSi as a secure web application that automates clinical data analysis and report generation using PHP, My-SQL, and HTML/JavaScript
- Shiny tool to score and rank peer reviewed publication in real time to support clinical analysis in NeGeSi. [Demo](#)
- Developed interactive data analytical plots for tracking and assessing the quality of runs, future predictions, and querying patterns within NeGeSi by integrating interactive libraries such as FusionCharts
- Extensively worked with tools such as R, Shiny on both Linux and Windows platform to tailor the needs for the NGS analysis pipeline such curating metadata, data wrangling and validating.
- Played a key role in data mining, curation, extracting FDA drugs and Clinical trial information relevant to the mutations noted in the VCF files.
- Worked with the IT team to create an in-house data warehousing solutions to store these data or build real time API calls to pull the data.
- Assisted in development of standard operating protocols and policies

RESEARCH EXPERIENCE

Research Associate (Part-time), Chemical and Life Sciences
Engineering, *Virginia Commonwealth University, VA- USA*

2015 Jun- 2016
Aug **1 year**

- Guided graduates and undergraduates projects by providing training in handling human embryonic stem cells and designing experiments
- Prepare the induced pluripotent cell lines from patient fibroblasts with mitochondrial disorders and extracted of DNA/ RNA to provide samples for Next generation sequencing towards a project goal to analyze the mitochondrial (mt) DNA Hetroplasmmy

- Budgeting, allocating grant funds create final reports to PI

Graduate Research Assistant, *Integrative Life Sciences Program*, 2009 - 2014
Virginia Commonwealth University, VA- USA **5 years**

- Isolation and characterization of lung stem / progenitor cells from p53 murine cancer models
 - Successfully established and characterized sub-populations of lung cells with stem cell characteristics from different murine p53 knock-in models
 - Applied parametric and non-parametric statistical methods, standard curve analysis using R statistical software as part of data analysis
 - Modeling gene networks using Ingenuity® pathway studio and R-Bioconductor packages

Laboratory Research Technician, *Virginia Institute of Psychiatry and Behavioral Genetics, Virginia Commonwealth University, VA- USA* 2012 - 2013
1 year

- Investigated the genetic and molecular mechanisms of gene expression in the development of psychiatry illness (Schizophrenia, Panic disorder)
 - Genotyping patient brain samples expressing variable number tandem repeats (VNTRs) of microRNA-137 precursor, molecular cloning
 - Gene expression and Sanger sequence analysis using R and sequence alignment tools

Student Research Assistant, *Chemical and Life Science Engineering*, 2007- 2009
Virginia Commonwealth University, VA- USA **2 years**

- Microarray data analysis
 - Worked with Affymetrix Human Genome U133A and Mouse exon expression arrays
 - Processing array data using Affymetrix power tools and packages in R programming
 - Applied statistical concepts of hierarchical clustering, regression models, multivariate analysis, and multiple hypotheses testing for data analysis
 - Extensively used R, MATLAB for data analysis and PERL scripting for data parsing

- Network modelling in R, Ingenuity®, and other open ware bioinformatics software

Research Internship, *National Chemical Laboratory, Pune, India* 2006 (6 mo)

- Genetic Algorithm & Support Vector Machine (GA-SVM) based prediction method for propensity of a protein to be soluble or to form inclusion body upon over-expression in Escherichia coli
 - Gathered published data on soluble and insoluble proteins, analyzed and synthesized data to generate random protein sequences. Coded GA-SVM algorithm in MATLAB to classify protein sequences based on composition of specific amino acid and their characteristics

EDUCATION

- **Ph.D. Integrative Life Sciences**, *Virginia Commonwealth University, USA* 2014
- **Masters in Bioinformatics**, *Virginia Commonwealth University, USA* 2009
- **Advanced diploma in Bioinformatics**, *University of Pune, Pune, India* 2007

TECHNOLOGY SUMMARY

- Bioinformatics: Working with large scale public data sets such as (TCGA data)
- Proficiency in R-programming, Shiny, Tableau, SQL, My-SQL, JavaScript, JQuery
- Familiarity with Python, Power BI , DOMO, SAS, PHP, AWS, Azure, GC working environment

HONORS AND AWARDS

- Awarded Department of Biotechnology, merit scholarship during the study period of Advanced Diploma in Bio-Informatics at University of Pune, Pune, India. [2005-2006]

PUBLICATIONS ([GOOGLE SCHOLAR LINK](#))